p-Adic Quantum Error Correction Classifier Verification: A Computational Methodology

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Abstract

The Number-Theoretic Ultrametric Foundations paper proposed three conjectures connecting p-adic valuation theory to quantum error-correcting (QEC) code classification: the CSS-Ultrametric Correspondence (C2.1'), Kodaira-Néron Fiber Classification for stabiliser codes (C5.1), and the Mahler v_p-Spectral Decomposition (C7.3'). Initial computational verification on 4 code families (83% accuracy) established these as theorem targets. This paper provides the computational methodology for full verification across the complete QEC landscape — an estimated 10^3 codes spanning 10+ families (surface, topological, colour, LDPC, concatenated, subsystem, CSS, stabiliser, GF(4), graph). The methodology specifies: (1) a code database compilation protocol from the QEC literature; (2) automated Mahler v_p spectrum computation using the existing QNFO Ultrametric Engine; (3) Kodaira-Néron fibre type classification via the /validate API; (4) statistical validation with bootstrapping and cross-validation. Success criteria: classification accuracy $\geq 60\%$ on a held-out test set of 50 code families (random baseline: ~25%). The methodology requires ~5 months of computational work and ~$40K in cloud resources. All tools are deployed on Cloudflare Workers with D1, R2, and Vectorize bindings. This paper is a computational methodology design — it does not perform the verification; it provides the roadmap. [SPECULATIVE]

Keywords: p-adic valuation, quantum error correction, stabilizer codes, Mahler spectrum, Kodaira-Néron classification, computational verification, ultrametric, Cloudflare Workers

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1. Background

1.1 The Conjectures

The Number-Theoretic Ultrametric Foundations paper [1, established — QNFO 2026] advanced three major conjectures connecting p-adic number theory to quantum error-correcting codes:

C2.1' (CSS-Ultrametric Correspondence). Every CSS code corresponds to an ultrametric tree. Specifically: the CSS construction from classical codes C_1, C_2 yields a metric on the logical subspace that is ultrametric if and only if C_1 and C_2 satisfy a mutual orthogonality condition in the p-adic valuation.

C5.1 (Kodaira-Néron Fiber Classification). The classification of stabiliser code types — CSS, GF(4), stabiliser, graph — corresponds to Kodaira-Néron fibre types (I_n, II, III, IV, I_0, *I_n*, IV, *III*, II*) from the theory of elliptic surfaces. Each code type maps to a specific fibre type based on its p-adic Mahler spectral signature.

C7.3' (Mahler v_p-Spectral Decomposition). The Mahler spectral analysis of a stabiliser code — computing the v_p-Mahler spectrum of the code's weight enumerator — distinguishes optimal codes (with v_p^max ≥ 28) from random ensembles (v_p^max ≤ 4). Optimal and random codes satisfy all three conjectures.

1.2 Initial Verification

The initial verification [1] covered 4 code families with 83% classification accuracy and 100% lemma-level agreement. The Mahler spectral analysis yielded v_p^max = 28 for optimal codes versus v_p^max = 4 for random ensembles. All three conjectures were satisfied (3/3) for both optimal and random codes.

However, 4 code families out of the estimated 10+ families in the QEC landscape is a small sample. Full verification requires expanding to all known code families and testing generalisation.

1.3 This Paper

This paper provides the computational methodology for full verification. It does NOT perform the verification — it is a design document, analogous to the G-07 DDG design in the Open Problems paper [2]. The methodology is specified at the level of API calls, data structures, and statistical tests, such that an implementation team could execute it without further design decisions.

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2. Methodology

2.1 Code Database Compilation

Source: The QEC literature since Calderbank-Shor-Steane (1996) contains $> 10^3$ distinct stabiliser codes across > 10 families.

Protocol: 1. Compile a comprehensive list of code families: surface codes, toric codes, colour codes, low-density parity-check (LDPC) codes, concatenated codes, subsystem codes, CSS codes, GF(4)-additive codes, stabiliser codes, graph codes, hypergraph product codes, quantum expander codes, homological codes, and anyonic codes. 2. For each family, select 10 representative codes with varying parameters ([[n, k, d]] with n from 5 to 1000, k from 1 to 100, d from 3 to 50). 3. Record metadata: code family, parameters [[n, k, d]], construction method, stabiliser generator matrix, logical operators, and weight enumerator. 4. Store in the QNFO D1 database with schema: `qec_codes (id, family, n, k, d, stabilizer_json, weight_enumerator, classification, verification_status)`.

Estimated effort: 1 month (literature survey + database population).

2.2 Mahler v_p Spectrum Computation

Tool: QNFO Ultrametric Engine [3, established — QNFO 2026], endpoint `/spectral-analysis`.

Protocol: 1. For each code in the database, extract the weight enumerator polynomial A(z). 2. Compute the p-adic Mahler coefficients c_k via the Mahler expansion: A(z) = Σ c_k · binomial(z, k). 3. Compute the v_p-Mahler spectrum: for each prime p dividing the code parameters n, compute the p-adic valuations v_p(c_k) of the Mahler coefficients. 4. Extract the maximum valuation v_p^max = max_k v_p(c_k) as the spectral signature. 5. Store results with schema: `qec_spectra (code_id, prime_p, vp_max, mahler_coeffs_json, computation_timestamp)`.

API call:

```
POST /spectral-analysis
Body: {"code_id": "...", "prime_p": 2, "method": "mahler"}
Response: {"vp_max": 28, "coefficients": [...], "status": "ok"}
```

Estimated effort: 2 months (API integration + batch computation for 10^3 codes). Parallelisation: the Ultrametric Engine is deployed on Cloudflare Workers at the edge, scaling to 10^3 concurrent requests.

2.3 Kodaira-Néron Fibre Type Classification

Tool: QNFO Ultrametric Engine [3], endpoint `/validate`.

Protocol: 1. For each code, submit the Mahler spectrum (computed in §2.2) to the `/validate` endpoint. 2. The endpoint classifies the code's Kodaira-Néron fibre type based on the v_p-Mahler spectral signature. 3. Compare the classified fibre type to the known code family: CSS → I_n, GF(4) → II, stabiliser → III, graph → IV, etc. (per C5.1). 4. Compute per-family and overall classification accuracy.

API call:

```
POST /validate
Body: {"spectrum": {...}, "method": "kodaira-neron"}
Response: {"fibre_type": "I_5", "confidence": 0.89, "classified_family": "CSS"}
```

Estimated effort: 1 month (API integration, classification for 10^3 codes).

2.4 Statistical Validation

Protocol: 1. **Stratified train/test split:** Hold out 50 code families (stratified across all known code types — CSS, GF(4), stabiliser, graph, LDPC, surface, etc.) for testing. Train the classifier on the remaining ~10^3 codes. 2. **Bootstrapping:** Resample the training data with replacement 10^3 times and compute the mean and 95% confidence interval of the classification accuracy. 3. **Cross-validation:** 10-fold cross-validation on the training set; report mean accuracy and standard deviation. 4. **Permutation test:** Shuffle the code-to-family mapping and re-run classification 10^3 times to establish the null distribution. A p-value < 0.01 indicates the classification is significantly better than random. 5. **Confusion matrix:** Report which code families are confused with which other families, providing insight into the structure of the ultrametric classification.

Estimated effort: 1 month (statistical analysis + visualisation).

2.5 Total Timeline

Phase	Timeline	Deliverable	Gate
Code database compilation	Month 1	D1 database with 10^3 codes	$\geq$ 100 codes in $\geq$ 10 families
Mahler spectrum computation	Months 2-3	v__p-Mahler spectra for all codes	$\leq$ 1% API failure rate
Fibre type classification	Month 4	Classification results	$\geq$ 80% accuracy on training
Statistical validation	Month 5	Validation report	Bootstrapped CI, $p < 0.01$

Total: 5 months. Estimated cost: ~$40K.

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3. Infrastructure

3.1 Cloudflare Integration

All components run on Cloudflare's edge network:

Component	Technology	Binding
Code database	Cloudflare D1 (SQLite)	`PAPERS_DB` → `living-paper`
Spectral analysis	Ultrametric Engine Worker	`/spectral-analysis`
Classification	Ultrametric Engine Worker	`/validate`
Results storage	Cloudflare R2	`qnfo/qec-verification/`
Semantic search	Cloudflare Vectorize	`qwav-research-v2`

3.2 Reproducibility

All computations are deterministic given the same input. The Docker container for the Ultrametric Engine is pinned to a specific commit hash in the QNFO/qnfo-skills repository. Results are stored in R2 with timestamped keys for auditability.

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4. Success Criteria

4.1 Primary Gate

[UNTESTED] Classification accuracy $\geq$ 60% on a held-out test set of 50 code families. Random guessing achieves ~25% accuracy (1 in 4 fibre types). A result below 60% indicates the ultrametric classification captures signal but not enough to be practically useful.

4.2 Secondary Gates

- **C2.1':** The CSS-Ultrametric Correspondence is verified if $\geq$ 90% of CSS codes are correctly classified as type I__n fibre.

- **C5.1:** The Kodaira-Néron map is verified if per-family accuracy exceeds 80% for all families.
- **C7.3':** The v_p-Mahler spectral gap (v_p^max difference between optimal and random codes) exceeds 20 for all primes $p \geq 2$.

4.3 Falsifiability

The conjectures are falsified if: 1. Classification accuracy falls below 60% on held-out test data (all three conjectures) 2. The Mahler spectrum of random codes shows v_p^max ≥ 20 (C7.3') 3. Per-family accuracy for any single family falls below 50% (C5.1 constrained for that family)

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5. Connection to the Adelic Programme

This computational verification is the QEC-theoretic component of the broader Adelic Physics Programme [4]. The key structural connection:

- The p-adic valuations v_p distinguish optimal from random codes because optimal codes have structure that random ensembles lack — exactly analogous to the way $\alpha \approx 1/137$ (a specific geometric eigenvalue) differs from a random dimensionless coupling constant.
- The Kodaira-Néron fibre classification maps code families to algebraic-geometric types because stabiliser codes, like elliptic surfaces, are classified by their singular fibres — and the classification is ultrametric.
- The Mahler spectrum connects number theory (p-adic valuations) to quantum information (code performance) via a spectral expansion — the same bridge that Poisson summation provides between discrete sums and continuous integrals [5].

If the conjectures are verified on the full QEC landscape, the adelic programme gains a concrete computational success — the p-adic framework provides a classifier that outperforms random guessing by a factor of 2.4×, with specific structural predictions (fibre types, Mahler spectra) that are falsifiable and computationally verifiable.

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6. Conclusion: The Invitation

This paper provides the computational methodology for verifying the three ultrametric QEC conjectures across the complete quantum error-correcting code landscape. The tools exist: the QNFO Ultrametric Engine [3] with its 27+ API endpoints. The data exists: the QEC literature contains $> 10^3$ stabiliser codes. The question is clear: does p-adic valuation theory classify quantum error-correcting codes with accuracy $\geq 60\%$, significantly above the random baseline?

The answer awaits execution. This paper is the invitation.

Declarations

Funding: This research received no specific grant from any funding agency.

Conflicts of Interest: The author declares no conflicts of interest.

Ethics Approval: Not applicable.

Author Contributions: Single author — all contributions.

Data Availability: No experimental data were generated or analysed.

Code Availability: The Ultrametric Engine is deployed on Cloudflare Workers: `https://ultrametric-engine.q`
The methodology described here extends the existing /spectral-analysis and /validate endpoints.

Use of Artificial Intelligence: AI-assisted drafting was used for literature synthesis and prose refinement.

References

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